Kapil Adhithya N

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Summary

AI/ML developer with hands-on experience building **LLM and RAG applications**, **machine learning models**, and **REST APIs** in **Python**. Built a **natural language to SQL agent** with **LangChain**, **ChromaDB**, and **prompt engineering**, and developed **regression models** for insurance claim prediction through **feature engineering** and **hyperparameter tuning**. Currently at Turing AI, reviewing model responses to support **LLM fine-tuning**.

Technical Skills

Programming and Databases: Python, SQL, JavaScript, MySQL, MongoDB, ChromaDB, C/C++, Java

Machine Learning and Deep Learning: scikit-learn, PyTorch, TensorFlow, XGBoost, regression (Linear Regression, Random Forest), classification, clustering, feature engineering, hyperparameter tuning, model evaluation

Data Analysis: Pandas, NumPy, statistical analysis, correlation analysis, outlier detection, data visualization

Generative AI and NLP: LLM applications, retrieval-augmented generation (RAG), prompt engineering, LangChain, Hugging Face, Ollama, vector databases (ChromaDB), text-to-SQL, contextual memory, LLM fine-tuning support

APIs and Deployment: FastAPI, Node.js, Express, REST APIs, model deployment (API serving), WebSockets

Tools: Git, VS Code, React, Recharts, Tailwind CSS, SDLC

Work Experience

Data Analyst | Turing AI — Jul 2025 – Present

- Reviewed and corrected model responses for the Gemini AI Code project to support model fine-tuning, checking each output for accuracy and alignment with project objectives.
- Created visualization-focused datasets for a Google project.
- Handled machine learning, deep learning (PyTorch, TensorFlow), and automation tasks.

AI & ML Intern | IQube Innovation Center — Jan 2024 – Mar 2024

- Analyzed insurance claim data with predictive modeling, risk segmentation, and statistical evaluation to identify high-cost customer segments and key health risk factors.
- Improved claim prediction accuracy through correlation analysis, outlier detection, and feature engineering.
- Compared Linear Regression, Random Forest, and XGBoost models and tuned their hyperparameters.

Projects

Natural Language SQL Agent | *LangChain, Ollama, ChromaDB, FastAPI, MongoDB, React* — Aug 2025 – Oct 2025

- Built an LLM-powered system that converts natural language queries into optimized, executable SQL statements across multiple database environments, using LangChain with Ollama for LLM integration.
- Added retrieval-augmented generation (RAG) with ChromaDB and prompt engineering to guide SQL generation, and improved query accuracy using contextual memory stored in MongoDB.
- Served the system through FastAPI REST APIs, with a React, Recharts, and Tailwind CSS front end that shows table relationships and data insights through interactive charts, plus user authentication and access control.

AI Code Debugger | *LangChain, Hugging Face (StarCoder), Node.js, Express, React* — Feb 2025

- Built an LLM tool that debugs code snippets with Hugging Face's StarCoder, returning fixes and explanations.
- Wrote the LangChain prompt template and exposed the model via a Node.js/Express REST API with a React UI.

Education

Kumaraguru College of Technology — Dec 2021 – Apr 2025

B.Tech. in Information Technology | CGPA: 8.74/10

Coursework: Machine Learning, Data Mining, Data Structures and Algorithms, Databases, Discrete Mathematics

Certifications and Awards

- **IBM:** Data Analytics using Python
- **Best Outgoing Student:** All-Rounder of the Department, Information Technology (2021–2025)
- **Mahatma Gandhi Scholarship:** Awarded for academic excellence in the pre-final and final years